Hiten Dalmia

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Education

Chennai Mathematical Institute — Chennai, India
Master of Science in Theoretical Computer Science (On Full Scholarship) — *Aug. 2026 – Discontinued*

Kirori Mal College, University of Delhi — Delhi, India
Bachelor of Science (Honours with Research) in Mathematics, Minor in Economics — *Nov. 2022 – May 2026*

Experience

Software Engineer — Mar. 2023 – Present
Freelance & Fractal — *Remote*

- Delivered 10+ production software applications and automation systems for client businesses, specializing in full-stack architecture, real-time data ingestion pipelines, and RESTful API infrastructure.
- **Order Procurement & Management Solution** | *PostgreSQL, NodeJS, TypeScript, NextJS*
  - Designed a full-stack order portal using NextJS, ExpressJS, TypeScript, and PostgreSQL with session-based authentication and secure RESTful endpoints.
  - Built a dual-tier system with real-time inventory tracking for clients and an admin control suite supporting task dispatch, automated audit logs, and scheduled recurring orders.
  - Constructed an automated order ingestion pipeline parsing incoming WhatsApp message payloads into relational SQL records, eliminating manual data entry and cutting processing time by 80%.
  - Optimized PostgreSQL database schemas with B-tree indexing and foreign key integrity constraints to process 5,000+ daily order queries with under 50ms response latency.

Software Engineer — Oct. 2023 – Apr. 2025
Tagglabs Experimental Pvt. Ltd. (Contract) — *Gurugram, Haryana (Remote)*

- **Cross-platform AR App** | *C/C++, OpenCV, Flutter, Django* **(Live link)**
  - Architected a native C/C++ bridge using `dart:ffi` to port OpenCV feature-matching algorithms into Flutter, enabling unified cross-platform image processing across Android and iOS.
  - Engineered a real-time frame processing pipeline normalizing OS video buffers and aspect ratios into raw C structs with zero-copy memory management and 30+ FPS throughput.
  - Implemented asynchronous feature extraction and object detection routines, transmitting coordinate payloads to Dart for real-time overlay rendering without UI thread blocking.
  - Developed responsive Flutter user interfaces and RESTful Django backend APIs to manage location-aware spatial telemetry and client discovery services for alpha release.
- **Kia National Conference App** | *Flutter*
  - Deployed an event management mobile app in Flutter for 500+ attendees at the Kia Conference in Bangkok, supporting real-time registration, attendee directories, and venue navigation.

Technical Skills

Systems & Languages: Python, C++, C, SQL, JavaScript/TypeScript, Dart, R
Automation & Infrastructure: Linux, Bash, Docker, AWS, RESTful APIs, Git, PostgreSQL
Frameworks & Databases: NodeJS, ExpressJS, Django, NextJS, Flutter, React Native, NestJS
Data Science & Computer Vision: OpenCV (SIFT), NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn

Achievements

Erasmus Mundus Joint Master MATHS-DISC Fellowship — Europe
Issued by European Union and affiliated universities — *Mar. 2026*

Master in Mathematics (ISI) and MSc Computer Science (CMI) Scholarships — India
Indian Statistical Institute and Chennai Mathematical Institute respectively — *Jun. 2026*

Publications

Computable Analysis: Mathematical Foundations and Computability of Real Functions
Poincare Journal of Mathematical Physics & Applications, Vol. 1, No. 1 (2026), 1–20 (with Sujit Kumar)